Complex Numbers: Equations with Complex Coefficients

Questions 1–14

Question 1. Find an equation of the form

$$f(z) = z^2 + pz + q$$

where p and q are complex numbers, given each of the following pairs of roots:

a) $z = 1 + i$ and $z = 3 + 2i$
b) $z = -3 + 4i$ and $z = 1 + 5i$
c) $z = -2 + 2i$ and $z = 2 - 2i$

Question 2. Find an equation of the form

$$f(z) = z^3 + pz^2 + qz + r$$

where p, q and r are complex numbers, given that the solutions to $f(z) = 0$ are $z = 1 + i$, $z = -1 + i$ and $z = 2 + 2i$

Question 3. A polynomial is defined by

$$f(z) = z^3 - (5+6i)z^2 + (35+27i)z - 58 - 30i$$

a) Find $(1+9i)^3$

b) Hence show that $z = 1+9i$ is a solution to $f(z) = 0$

It is known that a second solution to $f(z) = 0$ is $z = 2$

c) Find the remaining solution, writing your answer in the form $a+bi$ where a and b are integers to be found.

Question 4. Given that

$$f(z) = z^2 - (2+3i)z - k$$

where k is a complex number and $1+2i$ is a root of the equation $f(z) = 0$

a) Find the value of k

b) Show that $1-2i$ is not a root of the equation $f(z) = 0$

Question 5. It is known that $2 - i$ is a root of

$$f(z) = z^3 - 3iz^2 + 2z + k$$

where k is a complex number

a) Find the value of k

b) Show that $2 + i$ is not a solution to $f(z) = 0$

Question 6.

a) Find $f(z)$ such that

$$z^3 - i = (z + i)f(z)$$

b) Hence find all the solutions to $z^3 - i = 0$

Question 7. Find the solutions to

$$z^2 - (8 - 5i)z + 3 - 29i = 0$$

Question 8. The following questions should be solved by using methods relating the coefficients of a polynomial to sums and products of its roots

Given

$$z^2 - (10 - 2i)z + k = 0$$

where k is a complex number has roots a and $6 - 2i$. Find the value of k

Question 9. Given

$$z^2 - kz = -6 - 21i$$

where k is a complex number has roots a and $2 + 7i$. Find the value of k

Question 10. It is given that $1 + i$ and 3 are solutions of

$$z^3 - 2iz^2 + pz + q = 0$$

where p and q are complex numbers. Determine the values of p and q

Question 11. It is given that $2+i$ and -4 are solutions of

$$z^3 + pz^2 - (5+8i)z - q = 0$$

where p is a complex number and q is a real positive number. Find the values of p and q

Question 12. It is given that $2-3i$ and 3 are solutions of

$$2z^3 + pz^2 + qz - 13 = 0$$

where p and q are complex numbers. Determine the values of p and q

Question 13. The equation

$$z^3 - (10 - 2i)z^2 + pz + q = 0$$

where p and q are complex numbers has solutions $z = a$, $z = b$ and $z = a + bi$ for real values a and b

a) Find the values of a and b

b) Find the values of p and q

Question 14. Prerequisite knowledge for Q14: Argand Diagrams

The solutions of

$$z^3 - (6 + 3i)z^2 + (9 + 14i) - 15i = 0$$

all lie on the same straight line in the complex plane. It is known that two of the solutions are $z = 3$ and $z = 1 + 2i$
Find the remaining solution, writing your answer in the form $a + bi$ where a and b are integers to be found.